HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2012)

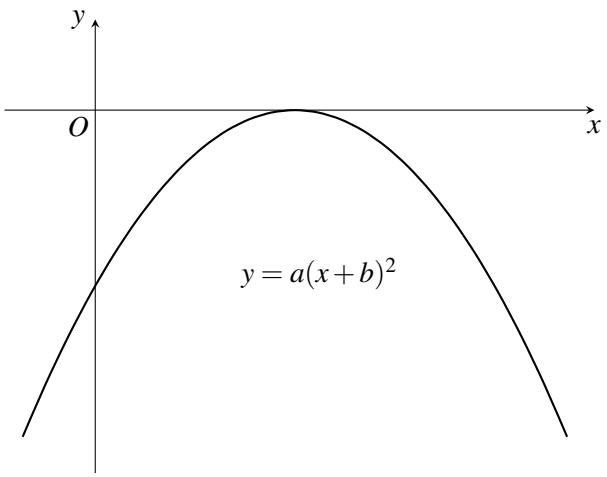
Student No.: 16 Name: _____ Marks: / 63

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The figure shows the graph of $y = a(x+b)^2$, where a and b are constants. Which of the following is true?



A. $a > 0$ and $b > 0$

B. $a > 0$ and $b < 0$

C. $a < 0$ and $b > 0$

D. $a < 0$ and $b < 0$

(DSE-2012 PII Q06)

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2. The length of a piece of thin string is measured as 25 m correct to the nearest m. If the string is cut into n pieces such that the length of each piece is measured as 5 cm correct to the nearest cm, find the greatest possible value of n.

A. 445

B. 566

C. 567

D. 650

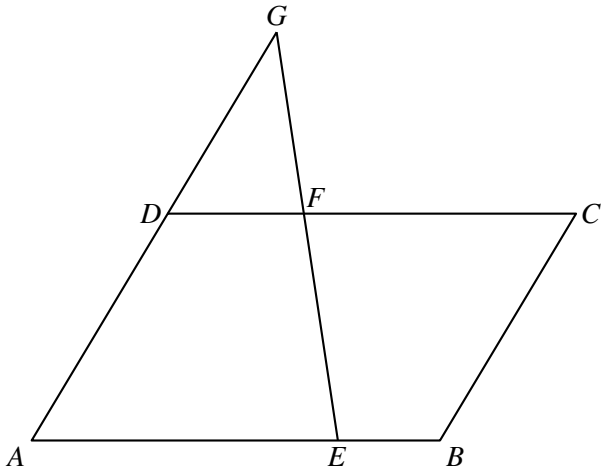
(DSE-2012 PII Q14)

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3. In the figure, $ABCD$ is a parallelogram. E and F are points lying on AB and CD respectively. AD produced and EF produced meet at G. It is given that $DF : FC = 3 : 4$ and $AD : DG = 1 : 1$. If the area of $\triangle DFG$ is 3 cm^2, then the area of the parallelogram $ABCD$ is



A. 12 cm^2.

B. 14 cm^2.

C. 18 cm^2.

D. 21 cm^2.

(DSE-2012 PII Q17)

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4. The rectangular coordinates of the point P are $(-3, -3\sqrt{3})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(3, 150°)$.

B. $(3, 330°)$.

C. $(6, 150°)$.

D. $(6, 330°)$.

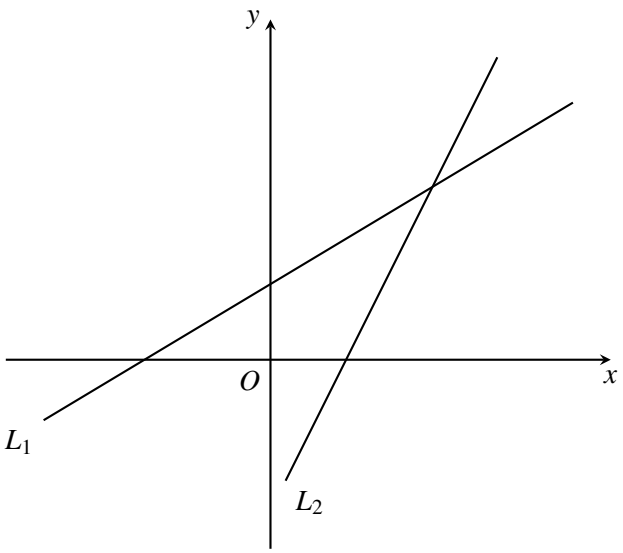
(DSE-2012 PII Q23)

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5. In the figure, the equations of the straight lines L_1 and L_2 are $ax+y=b$ and $cx+y=d$ respectively. Which of the following are true?



I. $a<0$

II. $a<c$

III. $b>d$

IV. $ad>bc$

A. I, II and III only

B. I, II and IV only

C. I, III and IV only

D. II, III and IV only

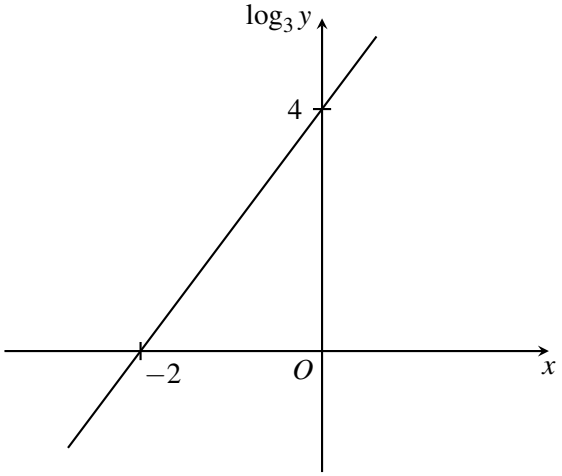
(DSE-2012 PII Q25)

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6. The graph in the figure shows the linear relation between x and $\log_3 y$. If $y = mn^x$, then $n =$



A. $\frac{1}{81}$.

B. $\frac{1}{9}$.

C. 9.

D. 81.

(DSE-2012 PII Q32)

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7. $AD000002012_{16} =$

A. $(10)16^{11} + (13)16^{10} + 8210.$

B. $(10)16^{12} + (13)16^{11} + 131360.$

C. $(11)16^{11} + (14)16^{10} + 8210.$

D. $(11)16^{12} + (14)16^{11} + 131360.$

(DSE-2012 PII Q33)

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8. A box contains six balls numbered 7, 8, 8, 9, 9 and 9 respectively. John repeats drawing one ball at a time randomly from the box without replacement until the number drawn is 9. Find the probability that he needs exactly three draws.

A. $\frac{1}{2}$

B. $\frac{1}{6}$

C. $\frac{1}{8}$

D. $\frac{3}{20}$

(DSE-2012 PII Q44)

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9. Make a the subject of the formula $\frac{3a+b}{8} = b - 1$. (3 marks) (DSE-2012 PI Q02)

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10. There are 132 guards in an exhibition centre consisting of 6 zones. Each zone has the same number of guards. In each zone, there are 4 more female guards than male guards. Find the number of male guards in the exhibition centre. (4 marks) (DSE-2012 PI Q05)

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11. (a) Find the value of k such that $x-2$ is a factor of $kx^3-21x^2+24x-4$.

(b) Figure 4 shows the graph of $y=15x^2-63x+72$. Q is a variable point on the graph in the first quadrant. P and R are the feet of the perpendiculars from Q to the x-axis and the y-axis respectively.

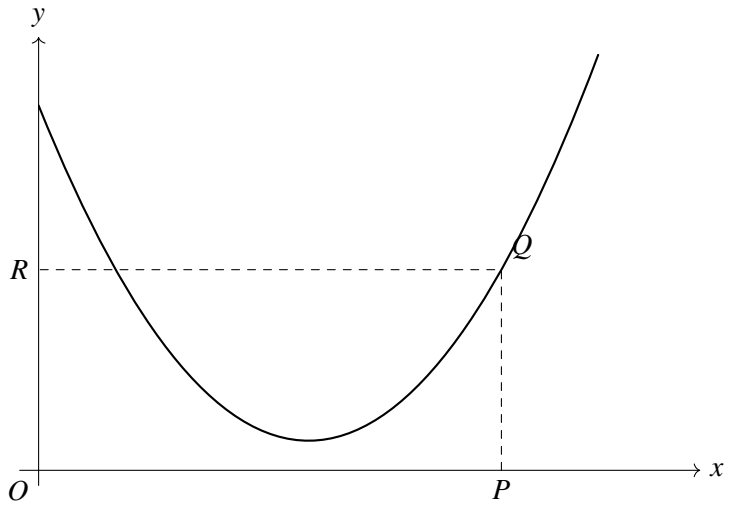


Figure 4

(i) Let $(m,0)$ be the coordinates of P. Express the area of the rectangle $OPQR$ in terms of m.

(ii) Are there three different positions of Q such that the area of the rectangle $OPQR$ is 12? Explain your answer.

(7 marks) (DSE-2012 PI Q13)

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12. The y-intercepts of two parallel lines L and ℓ are -1 and -3 respectively and the x-intercept of L is 3. P is a moving point in the rectangular coordinate plane such that the perpendicular distance from P to L is equal to the perpendicular distance from P to ℓ. Denote the locus of P by Γ.

(a) (i) Describe the geometric relationship between Γ and L.

(ii) Find the equation of Γ.

(b) The equation of the circle C is $(x-6)^2+y^2=4$. Denote the centre of C by Q.

(i) Does Γ pass through Q? Explain your answer.

(ii) If L cuts C at A and B while Γ cuts C at H and K, find the ratio of the area of $\triangle AQH$ to the area of $\triangle BQK$.

(9 marks) (DSE-2012 PI Q14)

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13. The standard deviation of the test scores obtained by a class of students in a Mathematics test is 10 marks. All the students fail in the test, so the test score of each student is adjusted such that each score is increased by 20% and then extra 5 marks are added.

(a) Find the standard deviation of the test scores after the score adjustment.

(b) Is there any change in the standard score of each student due to the score adjustment? Explain your answer.

(3 marks) (DSE-2012 PI Q15)

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14. Figure 5(a) shows a right pyramid $VABCD$ with a square base, where $\angle VAB = 72°$. The length of a side of the base is 20 cm. Let P and Q be the points lying on VA and VD respectively such that PQ is parallel to BC and $\angle PBA = 60°$. A geometric model is made by cutting off the pyramid $VPBCQ$ from $VABCD$ as shown in Figure 5(b).

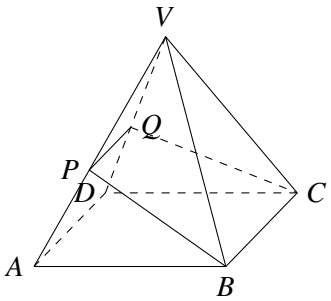


Figure 5(a)

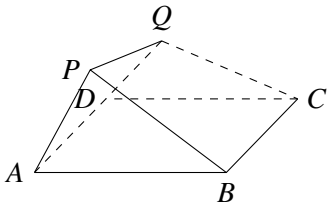


Figure 5(b)

(a) Find the length of AP.

(b) Let α be the angle between the plane $PBCQ$ and the base $ABCD$.

(i) Find α.

(ii) Let β be the angle between PB and the base $ABCD$. Which one of α and β is greater? Explain your answer.

(8 marks) (DSE-2012 PI Q18)

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15. In a city, the air cargo terminal X of an airport handles goods of weight $A(n)$ tonnes in the nth year since the start of its operation, where n is a positive integer. It is given that $A(n) = ab^{2n}$, where a and b are positive constants. It is found that the weights of the goods handled by X in the 1st year and the 2nd year since the start of its operation are 254 100 tonnes and 307 461 tonnes respectively.

(a) (i) Find a and b. Hence find the weight of the goods handled by X in the 4th year since the start of its operation.

(ii) Express, in terms of n, the total weight of the goods handled by X in the first n years since the start of its operation.

(b) The air cargo terminal Y starts to operate since X has been operated for 4 years. Let $B(m)$ tonnes be the weight of the goods handled by Y in the mth year since the start of its operation, where m is a positive integer. It is given that $B(m) = 2ab^m$.

(i) The manager of the airport claims that after Y has been operated, the weight of the goods handled by Y is less than that handled by X in each year. Do you agree? Explain your answer.

(ii) The supervisor of the airport thinks that when the total weight of the goods handled by X and Y since the start of the operation of X exceeds 20 000 000 tonnes, new facilities should be installed to maintain the efficiency of the air cargo terminals. According to the supervisor, in which year since the start of the operation of X should the new facilities be installed?

(13 marks) (DSE-2012 PI Q19)

I can finish _____ questions

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